EX-08 Reproducing an Image Using Prompts for Image Generation

Name:R.Divyadharshini

Register Number:212225230062

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slot:T2-H16

Aim

To demonstrate the ability of text-to-image generation tools to reproduce an existing image by crafting precise prompts.

Objective

To identify key visual elements in a reference image and reproduce a similar image using AI image generation tools through effective prompt engineering techniques.

AI Tools Used

1. ChatGPT Image Generation
2. Gemini
3. Bing Image Creator

Reference Image Selected

A realistic modern workspace image containing:

- Silver laptop
- Wooden desk
- White coffee mug
- Soft warm lighting
- Minimal background

Figure 1: Original Reference Image

Image Analysis

Element	Description
Objects	Laptop, coffee mug, notebook, desk
Background	Minimal modern workspace
Lighting	Soft warm natural lighting
Style	Realistic photography

Color Combination	Brown, white, silver
Camera Angle	Front-side angle

Algorithm

Step 1: Select the Reference Image

A modern workspace image with a laptop was selected as the reference image.

Step 2: Analyze the Image

The important visual elements such as objects, lighting, background, style, and camera angle were identified.

Step 3: Create Initial Prompt

A basic prompt describing the image was created.

Step 4: Generate Initial Output

The prompt was given to the AI image generation tool to generate the first output image.

Step 5: Refine the Prompt

Additional details regarding lighting, realism, texture, and composition were added to improve image quality.

Step 6: Generate Final Output

The refined prompt was used to generate a more accurate image.

Step 7: Compare the Results

The generated image was compared with the original reference image to evaluate similarity and improvements.

Initial Prompt

A laptop on a wooden desk with a coffee mug beside it, modern workspace, realistic style.

Initial Output

The AI tool generated an image containing the laptop and desk setup, but some details such as lighting and realism were limited.

Figure 2: Initial Generated Image

Refined Prompt

A realistic modern workspace with a silver laptop placed on a wooden desk, a white coffee mug beside the laptop, soft warm natural lighting coming from the side, minimal background, ultra realistic photography style, depth of field, cinematic composition, high detail, front-side camera angle.

Final Output

The generated image closely matched the original reference image with improved lighting, realistic textures, and better object placement.

Figure 3: Final Generated Image

Comparison Report

Feature	Original Image	AI Generated Image
Laptop Position	Centered on desk	Similar position achieved
Coffee Mug	Beside laptop	Correctly reproduced
Lighting	Soft warm lighting	Similar lighting achieved
Background	Minimal workspace	Closely matched
Realism	High	High
Overall Similarity	—	Very Similar

Figure 4: Comparison Between Original and Generated Images

4. COMPARISON: ORIGINAL vs FINAL GENERATED IMAGE

ORIGINAL IMAGE	FINAL GENERATED IMAGE
	

Feature	Original Image	Final Generated Image
Objects	Laptop, coffee mug, notebook, plant, books	Same objects accurately reproduced
Lighting	Soft warm natural light from side	Similar soft warm natural light achieved
Background	Minimal, clean workspace	Minimal, clean workspace
Color Combination	Brown, white, silver, green	Brown, white, silver, green
Camera Angle	Front-side angle	Front-side angle
Overall Similarity	–	**Very High Similarity**

Output

The AI image generation tool successfully reproduced an image similar to the original reference image using prompt engineering techniques.

Result

Thus, the original image was successfully reproduced using prompt engineering techniques and AI image generation tools.

References

1. OpenAI ChatGPT
2. Google Gemini
3. Bing Image Creator